Pseudocode

```
START
DISPLAY welcome message and title

REPEAT  // returns here if user chose N

  REPEAT
    DISPLAY "STEP 1: Choose starting base (2, 8, 10, 16)"
    INPUT inputbase
    IF inputbase is not 2, 8, 10, or 16 THEN
      DISPLAY "Invalid choice! Please try again."
    END IF
  UNTIL inputbase is 2, 8, 10, or 16
  SET frombase, fromname, and guidenum based on inputbase

  REPEAT
    DISPLAY "STEP 2: Choose target base (2, 8, 10, 16)"
    INPUT inputbase
    IF inputbase is not 2, 8, 10, or 16 THEN
      DISPLAY "Invalid choice! Please try again."
    END IF
  UNTIL inputbase is 2, 8, 10, or 16
  SET choice = "R"
  SET tobase and toname based on inputbase

  REPEAT  // returns here when the user chooses R

    // number inputting section
    REPEAT
      DISPLAY "Enter a" fromname "number"
      DISPLAY "Use only" guidenum
      INPUT number
      IF number is empty, has a digit not allowed in frombase, or is too big THEN
        DISPLAY "Invalid number! Please try again."
      END IF
    UNTIL number is valid

    // convert input to decimal section
    PROCESS number into decimal
    IF frombase is not 10 THEN
      DISPLAY conversion-to-decimal process (digit x base^position)
    END IF
    // decimal ready

    // Convert decimal to target base
    IF tobase = 10 THEN
      SET result = decimal
    ELSE
      PROCESS decimal into result by dividing by tobase and collecting the remainders
      DISPLAY decimal-to-target-base process
    END IF

    DISPLAY result

    // choosing next action
    REPEAT
      DISPLAY "R - Retry with same bases"
      DISPLAY "N - Choose new bases"
      DISPLAY "Q - Quit"
      INPUT choice
      IF choice is not R, N, or Q THEN
        DISPLAY "Invalid choice! Please type R, N, or Q."
      END IF

          UNTIL choice is R, N, or Q

        UNTIL choice is not "R"

    UNTIL choice = "Q"

    DISPLAY goodbye message
    END
```